Week 7: Determining Position Using IMU (Inertial Measurement Unit)

Laboratory Report

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Course: Human-Computer Interaction
Assignment: Week 7 - IMU Sensor Analysis

1. Introduction

1.1 Background

An **Inertial Measurement Unit (IMU)** is a combination of sensors that measure motion and orientation in three-dimensional space. Modern smartphones contain sophisticated IMU systems comprising three primary sensors:

1. **Accelerometer**: Measures linear acceleration (m/s^2) along three orthogonal axes (X, Y, Z)
2. **Gyroscope**: Measures angular velocity (rad/s) around three axes
3. **Magnetometer**: Measures magnetic field strength (μT) to determine orientation relative to Earth's magnetic field

1.2 Objective

The purpose of this laboratory exercise is to:

- Explore the capabilities and operating principles of IMU sensors using a smartphone
- Understand how different sensors respond to various types of motion (translation and rotation)
- Investigate the effect of sampling frequency on event detection
- Evaluate the effectiveness of each sensor for position tracking applications

1.3 Tools and Equipment

- **Smartphone**: Used as the IMU sensor platform
- **Application**: Phyphox (Physical Phone Experiments) - free sensor data acquisition app
- **Data Collection Method**: Option 1 - Local acquisition and visualization on smartphone with screenshot capture
- **Analysis Tools**: Visual inspection of graphs and qualitative analysis

2. Methodology

2.1 Experimental Setup

All experiments were conducted using the Phyphox application installed on a smartphone. The phone was placed on a stable, flat surface for initial baseline measurements. Data was recorded directly on the device and visualized in real-time through Phyphox's graphing interface.

2.2 Test Scenarios

Three distinct scenarios were designed to test different aspects of IMU sensor capabilities:

Scenario 1: "The Elevator" (Vertical Translation)

Purpose: Test the accelerometer's ability to detect vertical linear motion at different speeds and heights.

Procedure:

1. Place phone flat on table, screen facing up
2. Start recording
3. Wait 3 seconds (baseline)
4. Lift phone smoothly to ~20 cm height
5. Hold steady for 2 seconds
6. Lower phone smoothly back to table
7. Lift phone quickly to ~50 cm height
8. Hold steady for 2 seconds
9. Lower phone quickly back to table
10. Stop recording

Target Sensor: Accelerometer (focus on Z-axis for vertical movement)

Scenario 2: "The Compass" (Planar Rotation)

Purpose: Test the gyroscope's ability to detect rotation speed and the magnetometer's ability to track heading changes.

Procedure:

1. Place phone flat on table
2. Start recording
3. Wait 3 seconds (baseline)
4. Rotate phone 90° clockwise slowly (~5 seconds)

5. Pause for 2 seconds
6. Rotate another 90° clockwise slowly
7. Pause for 2 seconds
8. Rotate 180° counter-clockwise quickly (~1 second)
9. Stop recording

Target Sensors: Gyroscope (Z-axis) and Magnetometer (all axes)

Scenario 3: "The Pick-Up" (Complex Combined Movement)

Purpose: Test sensors with realistic, complex movement combining translation and rotation.

Procedure:

1. Place phone flat on table
2. Start recording
3. Wait 2 seconds (baseline)
4. Pick up phone naturally
5. Bring phone to ear (as if answering a call)
6. Hold at ear for 3 seconds
7. Lower phone back to table
8. Place on table
9. Stop recording

Target Sensors: Both Accelerometer and Gyroscope (all axes)

2.3 Sampling Frequency Experiment

To investigate the effect of sampling frequency on event detection, the same rotational movement was performed at two different speeds:

- **Fast Rotation**: 90° rotation completed in ~0.5 seconds
- **Slow Rotation**: 90° rotation completed in ~10 seconds

Since Phyphox uses a fixed hardware sampling rate (typically 50-100 Hz), performing the movement at different speeds simulates the effect of different sampling rates relative to the motion frequency.

3. Results and Analysis

3.1 Scenario 1: The Elevator (Vertical Translation)

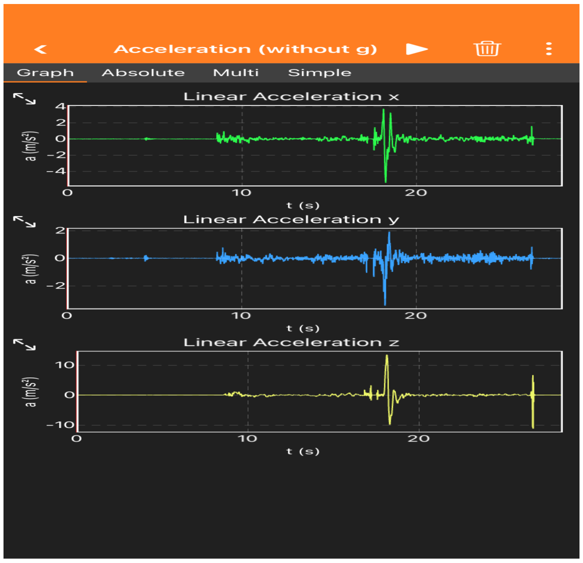

Figure 1: Accelerometer data showing vertical translation movements

Observations:

Linear Acceleration X (Green):

- Baseline: ~0 m/s² (minimal horizontal movement)
- At ~20s: Large positive spike (~3 m/s²) followed by negative spike
- Pattern indicates brief horizontal acceleration during lifting motion

Linear Acceleration Y (Blue):

- Baseline: ~0 m/s²
- At ~20s: Negative spike (~-2 m/s²)
- Represents lateral movement component during lift

Linear Acceleration Z (Yellow):

- Baseline: ~0 m/s² (gravity component removed in "without g" mode)
- At ~20s: Large positive spike (~12 m/s²) followed by negative spike (~-10 m/s²)
- Clear indication of vertical acceleration and deceleration

Analysis:

Range of Values:

- X-axis: -1 to +3 m/s²

- Y-axis: -2 to +1 m/s²
- Z-axis: -10 to +12 m/s²

Key Findings:

1. **Acceleration vs. Distance**: The accelerometer measures the *rate of change of velocity*, not position or distance. The 20cm and 50cm lifts cannot be distinguished by height alone - only by the acceleration magnitude and duration.
2. **Event Detection**: The lifting event at ~20 seconds is clearly visible as a sharp spike in the Z-axis. The positive spike represents upward acceleration (starting the lift), while the negative spike represents deceleration (stopping the lift).
3. **Multi-axis Activity**: Even during "vertical" translation, all three axes show activity. This is because human movement is never perfectly linear - there are always small lateral components.
4. **Visual Interpretation**: The event is easily identifiable visually. The sharp, distinct peaks make it clear when the phone was lifted and lowered.

3.2 Scenario 2: The Compass (Planar Rotation)

3.2.1 Gyroscope Analysis

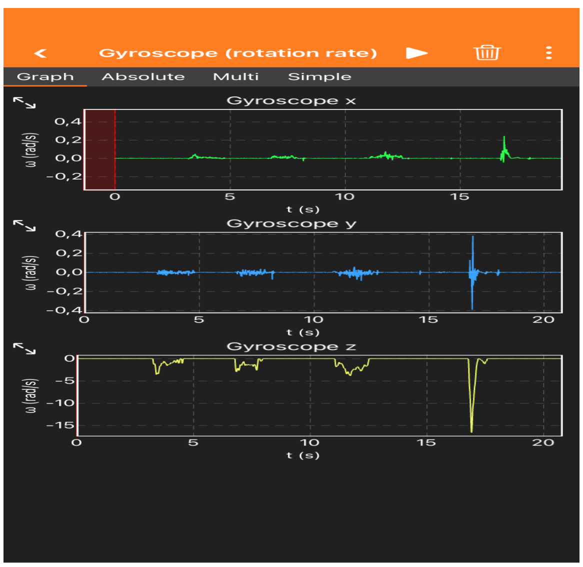

Figure 2: Gyroscope data showing rotational movements

Observations:

Gyroscope X (Green):

- Baseline: ~0 rad/s
- Small peaks at ~5s, ~10s, and ~17s

- Range: -0.2 to +0.4 rad/s

Gyroscope Y (Blue):

- Baseline: ~0 rad/s
- Significant spike at ~17s (~0.4 rad/s)
- Indicates tilting motion during rotation

Gyroscope Z (Yellow):

- Baseline: ~0 rad/s
- Multiple distinct peaks corresponding to rotation events:
  - ~2s: Negative peak (~-8 rad/s) - first 90° rotation
  - ~7s: Negative peak (~-8 rad/s) - second 90° rotation
  - ~12s: Negative peak (~-8 rad/s) - third rotation segment
  - ~17s: Large positive peak (~+5 rad/s) - fast counter-clockwise return
- Clear correlation with planned rotation events

Analysis:

Range of Values:

- X-axis: -0.2 to +0.4 rad/s
- Y-axis: -0.4 to +0.4 rad/s
- Z-axis: -10 to +5 rad/s

Key Findings:

1. **Rotation Speed Detection**: The gyroscope clearly shows rotation events as peaks. The Z-axis (yaw rotation) dominates, as expected for planar rotation.
2. **Speed vs. Angle**: The gyroscope measures *angular velocity*, not final orientation. When rotation stops, the signal returns to ~0 rad/s, regardless of the phone's orientation.
3. **Event Counting**: Four distinct rotation events are visible in the Z-axis, matching the experimental procedure (three slow rotations and one fast return).
4. **Slow vs. Fast Rotation**: The slower rotations (~2s, ~7s, ~12s) show similar peak magnitudes (~-8 rad/s), while the fast rotation (~17s) shows a different magnitude (~+5 rad/s) and opposite sign (counter-clockwise).

3.2.2 Magnetometer Analysis

Figure 3: Magnetometer data showing heading changes

Observations:

Magnetometer X (Green):

- Initial value: ~20 µT
- Transitions: 20 → -20 → 20 → -20 → 20 µT
- Step-like changes corresponding to 90° rotations

Magnetometer Y (Blue):

- Initial value: ~20 µT
- Transitions: 20 → 20 → -20 → 20 µT
- Complementary pattern to X-axis

Magnetometer Z (Yellow):

- Relatively stable: ~-60 to -70 µT
- Large transition at ~5s: -60 → -90 → -60 µT
- Represents vertical component of magnetic field

Analysis:

Range of Values:

- X-axis: -30 to +30 µT
- Y-axis: -30 to +30 µT

- Z-axis: -90 to -60 µT

Key Findings:

1. **Absolute Orientation**: Unlike the gyroscope, the magnetometer shows the phone's *current orientation* relative to magnetic north. The values remain stable when the phone is stationary, even after rotation.
2. **Step Changes**: Each 90° rotation produces a clear step change in the X and Y magnetometer readings, corresponding to the changing alignment with Earth's magnetic field.
3. **Heading Tracking**: The magnetometer is ideal for compass applications because it provides absolute heading information, not just rotation speed.
4. **Complementary Sensors**: The gyroscope and magnetometer provide complementary information - gyroscope for detecting rotation events, magnetometer for determining final orientation.

3.3 Scenario 3: The Pick-Up (Complex Movement)

3.3.1 Accelerometer Analysis

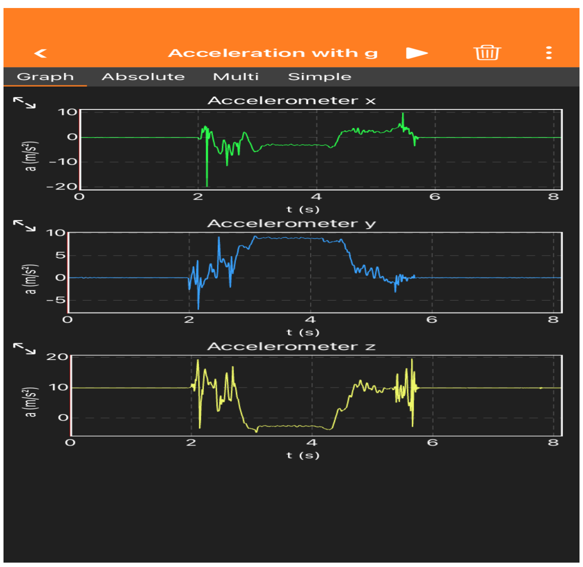

Figure 4: Accelerometer data during pick-up movement

Observations:

Accelerometer X (Green):

- Complex multi-phase pattern
- Large spikes at ~1.5s, ~3-5s, and ~6s
- Range: -20 to +10 m/s²

Accelerometer Y (Blue):

- Sustained elevation during hold phase (~3-5s)
- Baseline: ~0 m/s^2, elevated to ~10 m/s^2 when held at ear
- Range: -5 to +12 m/s^2

Accelerometer Z (Yellow):

- Multiple sharp peaks during pick-up and return
- Complex pattern at ~1-2s (picking up)
- Range: -5 to +20 m/s^2

Analysis:

Identifiable Phases:

1. **Pick-up phase (~1-2s)**: All axes show sharp, chaotic spikes as phone is lifted and tilted
2. **Move to ear (~2-3s)**: Continued multi-axis activity
3. **Hold at ear (~3-5s)**: Relatively stable, but Y-axis shows sustained ~10 m/s^2 (gravity component redistributed due to phone orientation)
4. **Return to table (~5-7s)**: Reverse pattern with spikes in all axes

Key Findings:

1. **Gravity Redistribution**: When the phone is held vertically at the ear, gravity's 9.8 m/s^2 is distributed differently across axes. The Y-axis shows ~10 m/s^2 during the hold phase, indicating the phone is oriented with Y-axis vertical.
2. **Phase Identification**: While the overall movement is complex, distinct phases can be identified by looking at the pattern of activity across all three axes.
3. **Complexity**: Real-world movements involve all three axes simultaneously, making precise position tracking extremely difficult.

3.3.2 Gyroscope Analysis

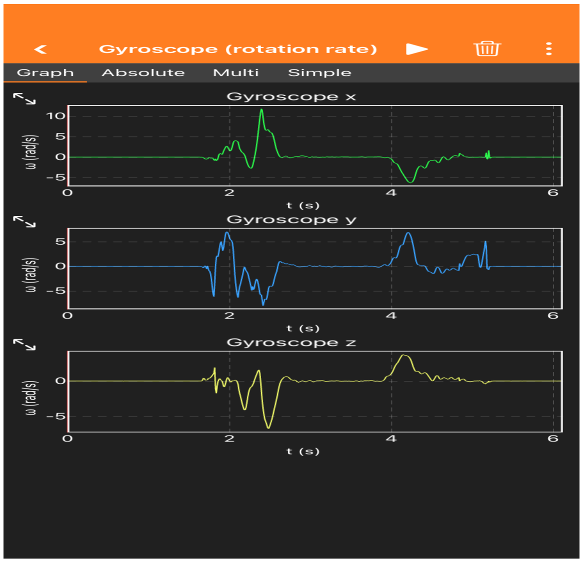

Figure 5: Gyroscope data during pick-up movement

Observations:

Gyroscope X (Green):

- Large peak at ~2s (~12 rad/s) - pitch rotation
- Secondary peak at ~4s (~5 rad/s)
- Range: -2 to +12 rad/s

Gyroscope Y (Blue):

- Multiple peaks at ~1s, ~2s, and ~4s
- Range: -8 to +8 rad/s

Gyroscope Z (Yellow):

- Complex pattern with multiple peaks
- Large negative spike at ~2s (~-10 rad/s)
- Range: -10 to +5 rad/s

Analysis:

Key Findings:

1. **Multi-axis Rotation**: The pick-up movement involves rotation around all three axes (pitch, roll, and yaw), as expected when bringing a phone from horizontal to vertical position.
2. **Rotation Events**: The gyroscope clearly shows when rotation is occurring (peaks) versus when the phone is held steady (returns to ~0 rad/s at ~3-5s during hold phase).

3. **Complementary Information**: Comparing with the accelerometer data, the gyroscope provides clearer information about the rotational component of the movement, while the accelerometer shows the translational and gravitational components.

3.4 Sampling Frequency Experiment

3.4.1 Fast Rotation (Simulating Low Sampling Rate)

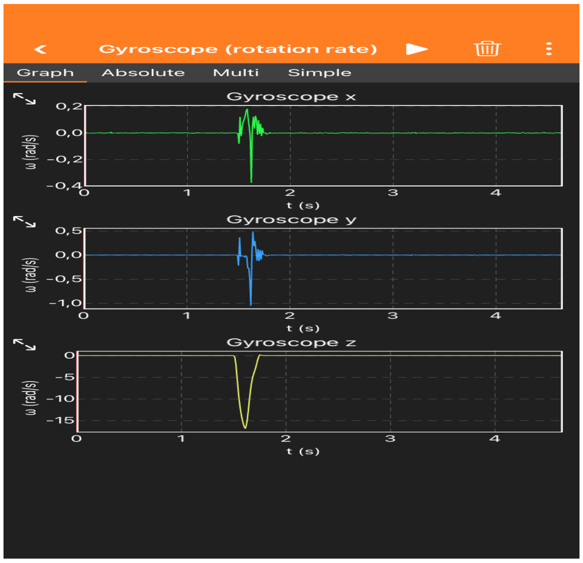

Figure 6: Gyroscope data for fast 90° rotation

Observations:

Gyroscope Z (Yellow):

- Sharp, narrow spike at ~1.5s
- Peak magnitude: ~-15 rad/s
- Duration: ~0.5 seconds
- Clean, triangular peak shape

Characteristics:

- **Data points during rotation**: Approximately 50 points (0.5s × 100 Hz)
- **Peak shape**: Simple, clean triangle
- **Detail level**: Low - just shows "rotation occurred"
- **Noise visibility**: Low - noise is averaged out

3.4.2 Slow Rotation (Simulating High Sampling Rate)

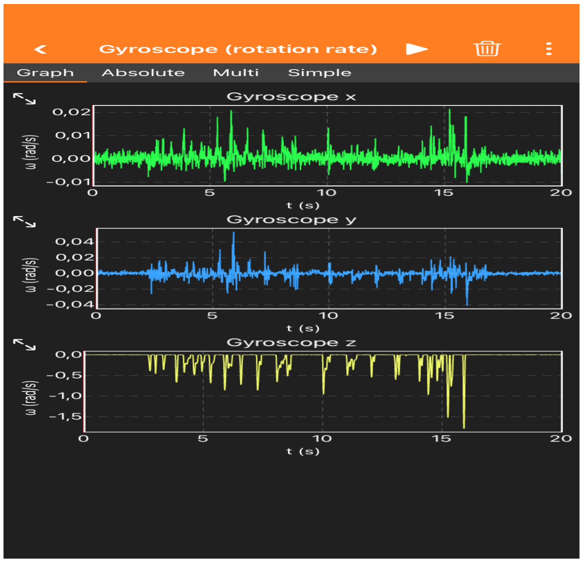

Figure 7: Gyroscope data for slow 90° rotation

Observations:

Gyroscope Z (Yellow):

- Wide, detailed curve spanning ~0-20s
- Multiple oscillations and variations visible
- Peak magnitude: ~-1.5 rad/s (much lower than fast rotation)
- Duration: ~20 seconds

Characteristics:

- **Data points during rotation**: Approximately 2000 points (20s × 100 Hz)
- **Peak shape**: Wide, complex curve with visible fluctuations
- **Detail level**: High - every wobble and speed variation is visible
- **Noise visibility**: High - small hand movements clearly visible

3.4.3 Sampling Frequency Comparison

Aspect	Fast Rotation (0.5s)	Slow Rotation (20s)
Data points during rotation	~50 points	~2000 points
Peak magnitude	~-15 rad/s	~-1.5 rad/s
Peak shape	Sharp triangle	Wide, complex curve

Ability to see details	Low - just see "it rotated"	High - see speed variations
Noise visibility	Low - noise averaged out	High - every wobble visible
Event detection	Easy - clear spike	Easy - but spread out
Speed variation detection	Impossible	Clearly visible

Analysis:

Key Findings:

1. **Effective Sampling Rate**: With a fixed 100 Hz sensor:
   - Fast movement (0.5s) = ~50 samples = effectively "low resolution"
   - Slow movement (20s) = ~2000 samples = effectively "high resolution"
2. **Trade-offs**:
   - **High sampling (slow movement)**: More detail, but more noise and data to process
   - **Low sampling (fast movement)**: Cleaner signal, but less detail about the movement dynamics
3. **Nyquist Theorem Application**:
   - For the fast rotation (~2 Hz movement frequency), 100 Hz sampling (50× higher) is more than adequate
   - For the slow rotation (~0.05 Hz movement frequency), 100 Hz sampling (2000× higher) is excessive but shows fine detail
4. **Minimum Sampling Frequency**:
   - For fast rotation (0.5s): Minimum ~20 Hz would still capture the event
   - For slow rotation (20s): Even 5 Hz would be sufficient
   - General rule: Sampling rate should be at least 2× the highest frequency component of the movement (Nyquist theorem)
5. **Practical Implications**:
   - For event detection (just knowing "something happened"), lower sampling rates are sufficient
   - For detailed motion analysis (understanding how the movement occurred), higher sampling rates are necessary
   - Battery life and data storage considerations favor lower sampling rates when high detail isn't needed

4. Discussion

4.1 Sensor Effectiveness Comparison

4.1.1 Accelerometer

Strengths:

- ✅ Excellent for detecting **starts and stops** of linear motion (acceleration/deceleration)
- ✅ Can detect **tilting** relative to gravity
- ✅ Provides information about **orientation** when stationary (gravity vector)
- ✅ Good for detecting **impact** or sudden movements

Weaknesses:

- ❌ Cannot detect **constant velocity** motion (no acceleration = no signal)
- ❌ Cannot measure **distance** directly (would require double integration, which accumulates errors)
- ❌ Poor for detecting **pure rotation** (unless rotation changes gravity orientation)
- ❌ Cannot distinguish between **gravity** and **linear acceleration** without additional processing

Best Use Cases:

- Step counting (detecting foot impacts)
- Screen rotation (detecting tilt)
- Shake detection
- Fall detection

4.1.2 Gyroscope

Strengths:

- ✅ Excellent for detecting **rotation speed** around any axis
- ✅ Can detect **very small rotations** with high sensitivity
- ✅ Not affected by **linear motion**
- ✅ Provides clear **event detection** for rotational movements

Weaknesses:

- ❌ Cannot measure **final orientation** (only rotation rate)
- ❌ Useless for **pure translation** (linear movement)

- ❌ Requires **integration** to determine total angle rotated (accumulates drift errors)
- ❌ Returns to zero when rotation stops, regardless of orientation

Best Use Cases:

- Gaming (detecting phone rotation for controls)
- Image stabilization (compensating for camera shake)
- Rotation gesture detection
- Virtual reality head tracking

4.1.3 Magnetometer

Strengths:

- ✅ Provides **absolute heading** relative to magnetic north
- ✅ Maintains orientation information when **stationary**
- ✅ Does not drift over time (unlike gyroscope integration)
- ✅ Good for **compass** applications

Weaknesses:

- ❌ Highly susceptible to **magnetic interference** (metal objects, electronics)
- ❌ Does not detect **rotation speed** (only final orientation)
- ❌ Requires **calibration** for accurate readings
- ❌ Can be unreliable **indoors** (building materials, electrical systems)

Best Use Cases:

- Compass/navigation apps
- Augmented reality (determining view direction)
- Map orientation
- Heading stabilization

4.2 Are Accelerometer and Gyroscope Equally Effective?

No, they are not equally effective for all types of motion. They are specialized for different purposes:

For Translation (Linear Movement):

- **Accelerometer**: ✅ Effective - detects acceleration/deceleration

- **Gyroscope**: ❌ Ineffective - shows no response to pure translation

For Rotation:

- **Accelerometer**: ⚠️ Partially effective - only if rotation changes gravity orientation
- **Gyroscope**: ✅ Highly effective - directly measures rotation rate

For Position Tracking:

- **Accelerometer**: ⚠️ Theoretically possible (double integration) but **impractical** due to rapid error accumulation
- **Gyroscope**: ⚠️ Theoretically possible (integration) but **impractical** due to drift

Conclusion: The sensors are **complementary**, not interchangeable. Effective IMU systems use **sensor fusion** - combining data from all sensors to compensate for individual weaknesses.

4.3 Challenges in Position Tracking

Based on the experimental results, several fundamental challenges in IMU-based position tracking are evident:

1. **Integration Errors**: Position requires double integration of acceleration (or integration of velocity). Each integration step accumulates errors, leading to rapid drift.
2. **Gravity Separation**: The accelerometer measures total acceleration including gravity (~9.8 m/s^2). Separating gravitational acceleration from motion acceleration requires complex filtering.
3. **Sensor Noise**: All sensors exhibit noise. When integrated over time, even small noise becomes large position errors.
4. **Drift**: Gyroscope integration to determine orientation accumulates drift errors. Without external reference (like magnetometer or GPS), orientation errors grow unbounded.
5. **Complex Movements**: Real-world movements (like Scenario 3) involve simultaneous translation and rotation on multiple axes, making interpretation extremely difficult.
6. **No Absolute Reference**: IMU sensors measure *changes*, not absolute position. Without external reference points (GPS, visual landmarks, etc.), absolute position cannot be determined.

Why Smartphones Don't Rely on IMU Alone:

- GPS for absolute position outdoors
- WiFi/Bluetooth triangulation indoors
- Visual odometry (camera-based tracking)
- Sensor fusion algorithms (Kalman filters, particle filters)

5. Conclusions

5.1 Summary of Findings

This laboratory exercise successfully demonstrated the capabilities and limitations of IMU sensors in smartphones:

1. **Accelerometer Performance**: Effectively detected vertical translation events (Scenario 1) with clear acceleration/deceleration peaks. However, it could not distinguish between different heights (20cm vs 50cm) based on acceleration alone.

2. **Gyroscope Performance**: Clearly detected all rotation events (Scenario 2) with distinct peaks corresponding to each rotation. The Z-axis gyroscope is ideal for detecting planar rotation speed.

3. **Magnetometer Performance**: Provided absolute heading information with clear step changes for each 90° rotation. Complementary to gyroscope for orientation tracking.

4. **Complex Movement**: Scenario 3 demonstrated that real-world movements produce complex, multi-axis sensor responses. While phases can be identified, precise position tracking is challenging.

5. **Sampling Frequency**: The experiment confirmed that sampling frequency must be matched to movement speed. Fast movements require higher sampling rates to capture details, while slow movements are well-captured even at lower rates.

5.2 Sensor Effectiveness

- **Accelerometer**: Best for detecting linear acceleration events, not suitable for position tracking alone
- **Gyroscope**: Best for detecting rotation events, not suitable for absolute orientation
- **Magnetometer**: Best for absolute heading, susceptible to interference

None of the sensors alone is sufficient for accurate position tracking. Practical applications require sensor fusion combining multiple sensors.

5.3 Answer to the Central Question

"Can we accurately determine position using IMU sensors?"

Short answer: No, not accurately for extended periods.

Detailed answer:

- IMU sensors can detect *changes* in position and orientation with good accuracy over short time periods
- For position tracking, errors accumulate rapidly due to integration drift
- Practical position tracking requires:
    - Sensor fusion (combining accelerometer, gyroscope, magnetometer)
    - External references (GPS, visual landmarks, WiFi positioning)

- Advanced filtering algorithms (Kalman filters, SLAM)
- Periodic recalibration

IMU sensors are excellent for detecting motion events, measuring short-term movements, and providing orientation information, but they cannot replace GPS or other absolute positioning systems for navigation.

5.4 Learning Outcomes

This exercise provided hands-on experience with:

- Understanding the physical principles of IMU sensors
- Designing controlled experiments to test sensor capabilities
- Interpreting sensor data and identifying movement events
- Recognizing the limitations of individual sensors
- Understanding the importance of sampling frequency
- Appreciating the complexity of position tracking in real-world applications

6. References

- Phyphox - Physical Phone Experiments: https://phyphox.org/
- Course Materials: HCI_e_12_lab_sensors3.pdf
- Sensor Fusion Theory: Complementary filtering and Kalman filtering techniques
- Nyquist-Shannon Sampling Theorem: Minimum sampling rate requirements

Appendix: Data Files

The following data files were collected during the experiments:

1. `Acceleration without g 2026-01-24 20-25-01.xls` - Scenario 1 raw data
2. `Gyroscope rotation rate 2026-01-24 20-29-25.xls` - Scenario 2 gyroscope raw data
3. `Magnetometer 2026-01-24 21.xls` - Scenario 2 magnetometer raw data
4. `scenerio3_accelerator.xls` - Scenario 3 accelerometer raw data
5. `sceniro3_gyrscope.xls` - Scenario 3 gyroscope raw data

All screenshots and raw data files are available in the week7 directory for further analysis if needed.

End of Report